QirbGarage Mobile Application

Frontend Architecture & Product Documentation
Flutter (Clean Architecture + BLoC)

1. Introduction

QirbGarage is a mobile platform designed to connect vehicle owners with nearby garages, spare part sellers, and repair manuals. The application helps users easily locate garages, request mechanics, discover spare parts, and learn vehicle repair techniques.

This document provides a structured overview of the mobile application from a frontend perspective. It defines the application modules, user flows, features, architecture, UI structure, and system behavior required for building the Flutter mobile app.

Scope of this document focuses only on the frontend mobile application built with Flutter. Backend services and infrastructure are considered external dependencies.

2. Objectives

The main goals of QirbGarage include:

- Help users quickly find nearby garages based on their location
- Allow garage owners to list and manage their services
- Enable a marketplace for spare parts exchange
- Provide a hub for automotive repair manuals
- Support communication between car owners and garages
- Deliver a scalable mobile experience using modern architecture

3. User Roles

The application supports two primary roles.

- **User (Vehicle Owner)**
  - Register and login
  - Detect location
  - Search nearby garages
  - Contact garages
  - Request mechanics
  - Browse spare parts
  - Upload or view manuals

- **Garage Owner**
  - Register garage account
  - Add and manage garage information
  - Accept mechanic requests
  - Post spare parts
  - Upload manuals
  - Communicate with users

4. Core Application Modules

The system contains three main modules.

1. Find Garage

   Allows users to discover garages near them.

2. Central Spare

   A marketplace for buying and selling spare parts.

3. Manual Hub

A knowledge sharing platform for repair manuals.

4.1. Find Garage Module

Purpose:

Allow users to locate nearby car repair garages.

Features:

- Location detection
- Garage listing
- Garage filtering by services
- Garage ratings
- Request mechanic
- Contact garage
- View services

Garage Information Displayed:

- Garage Name
- Distance
- Rating
- Availability status
- Service categories
- Contact button
- Request mechanic button

User Flow: Find Garage

User opens app

↓

Location detected

↓

User searches or filters garages

↓

List of garages displayed

↓

User selects a garage

↓

Garage details page

↓

User contacts garage OR requests mechanic

4.2. Central Spare Marketplace

Purpose:

Allow users and garages to buy or sell spare parts.

Key Features:

- Post spare part
- Search spare parts
- Filter spare parts
- Contact seller

- Identify part condition (new or used)

Two Primary Actions:

I Have This Part:

User posts a spare part for sale.

I Need This Part:

User requests a spare part they are looking for.

User Flow: Spare Part Marketplace

User opens Central Spare page

↓

User selects "I Need This Part" or "I Have This Part"

↓

User fills part details

↓

Part post created

↓

Other users search or browse parts

↓

User contacts seller

4.3. Manual Hub

Purpose:

Provide educational resources and repair documentation.

Content Types:

- PDF manuals
- Video tutorials
- Written guides

Categories:

- Engine
- Electrical
- Body
- General

Manual Types:

- Free manuals
- Premium manuals

Video Preview Feature:

Videos will appear in a short vertical scrolling interface Like TikTok to preview repair knowledge and attract users to full manuals.

User Flow: Manual Hub

User opens Manual Hub

↓

User searches manuals

↓

Filter by type or category

↓

User opens manual preview

↓

User downloads free manual OR unlocks premium content

5. Application Screen List

This section describes all screens included in the QirbGarage mobile application. Screens are organized based on system modules and user roles.

The application contains approximately **25–28 screens** across authentication, garage discovery, spare parts marketplace, manual hub, and profile management.

5.1 Authentication Screens

These screens allow users to create accounts and access the platform.

5.1.1 Splash Screen

Purpose:
Displayed when the application starts.

Functions:

- Shows the QirbGarage logo
- Initializes the application
- Checks user authentication status
- Redirects user to login or home screen

Key UI Components:

- App logo

- Loading indicator

5.1.2 Onboarding Screen (Optional)

Purpose:
Introduces new users to the application features.

Content:

- Discover Nearby Garages
- Buy and Sell Spare Parts
- Access Repair Manuals

Components:

- Feature illustrations
- Short description texts
- “Get Started” button

5.1.3 Login Screen

Purpose:
Allows existing users to access the platform.

Features:

- Email or phone login
- Password input
- Forgot password option
- Link to registration screen

UI Components:

- Email/Phone field
- Password field
- Login button

- Forgot password link
- Sign-up navigation

5.1.4 Register Screen

Purpose:
Allows new users to create accounts.

Required Fields:

- Full Name
- Email
- Phone Number
- Password
- Confirm Password

Actions:

- Register account
- Navigate to login

5.1.5 Role Selection Screen

Purpose:
Allows users to choose their account type.

Roles Available:

- Vehicle Owner (User)
- Garage Owner

This role determines the features available within the application.

5.2 Find Garage Module Screens

These screens support the main feature of discovering and interacting with garages.

5.2.1 Garage List Screen (Home Screen)

Purpose:
Displays nearby garages based on user location.

Features:

- Automatic location detection
- Garage search functionality
- Service category filters
- List of nearby garages

Displayed Information:

- Garage name
- Distance from user
- Rating
- Availability status

UI Components:

- Search bar
- Filter chips (Electrical, Mechanical, Bodywork)
- Garage cards
- Map icon

5.2.2 Garage Details Screen

Purpose:
Displays detailed information about a selected garage.

Information Displayed:

- Garage name
- Address and location
- Rating and reviews
- Services offered
- Working hours

- Contact information

Actions:

- Call garage
- Request mechanic
- View location on map

5.2.3 Request Mechanic Screen

Purpose:
Allows users to request a mechanic from a selected garage.

Form Fields:

- Service type
- Vehicle model
- Problem description
- Preferred service time

Action:

- Submit request to garage owner

5.2.4 Garage Reviews Screen

Purpose:
Displays user feedback about the garage.

Information Displayed:

- User ratings
- Written reviews
- Average rating score

Users can also submit their own reviews.

5.2.5 Garage Map Screen

Purpose:
Displays the garage location on a map.

Features:

- Interactive map
- Route directions
- Distance from current location

5.3 Central Spare Marketplace Screens

These screens enable spare parts trading between users and garages.

5.3.1 Spare Parts Feed Screen

Purpose:
Displays available spare parts posted by users and garages.

Features:

- Search spare parts
- Filter by condition (New / Used)
- Filter by vehicle model
- Filter by location

Displayed Information:

- Part name
- Vehicle model
- Price
- Seller name
- Location

5.3.2 Spare Part Details Screen

Purpose:
Displays detailed information about a spare part listing.

Information Displayed:

- Part name
- Description
- Vehicle compatibility
- Condition
- Price
- Seller contact

Actions:

- Contact seller
- View seller location

5.3.3 Create Spare Post Screen

Purpose:
Allows users to sell spare parts.

Fields:

- Part name
- Vehicle model
- Condition
- Description
- Price
- Location
- Image upload

Action:

- Publish spare part listing

5.3.4 Request Spare Part Screen

Purpose:
Allows users to request a spare part they need.

Fields:

- Part name
- Vehicle model
- Description
- Location

Other users or garages can contact the requester if they have the part.

5.3.5 My Spare Posts Screen

Purpose:
Displays spare parts posted by the user.

Features:

- View post status
- Edit post
- Delete post

5.4 Manual Hub Screens

These screens provide repair knowledge and documentation.

5.4.1 Manuals Feed Screen

Purpose:
Displays available repair manuals.

Filters:

- Engine

- Electrical
- Body
- General

Manual Types:

- PDF
- Video
- Guide

5.4.2 Manual Details Screen

Purpose:
Displays details of a selected manual.

Information Displayed:

- Manual title
- Vehicle model
- Author
- Number of pages or video duration
- Description
- Price (if premium)

Actions:

- Download manual
- Watch manual video

5.4.3 Video Manual Viewer Screen

Purpose:
Displays repair videos in a vertical scrolling format similar to short video platforms.

Features:

- Swipe to next video

- Preview manual tutorials
- Link to full manual content

5.4.4 Upload Manual Screen

Purpose:
Allows users and garage owners to upload repair manuals.

Fields:

- Manual title
- Vehicle model
- Category
- Description
- File upload (PDF / Video)
- Free or Premium selection

5.4.5 My Manuals Screen

Purpose:
Displays manuals uploaded by the user.

Actions:

- Edit manual
- Delete manual
- View download statistics

5.5 Profile and Account Screens

These screens allow users to manage their personal accounts.

5.5.1 Profile Screen

Purpose:
Displays the user's account information.

Information Displayed:

- Name
- Email
- Phone number
- Role type

Options:

- Edit profile
- View personal posts
- Logout

5.5.2 Edit Profile Screen

Purpose:
Allows users to update their account details.

Fields:

- Name
- Phone number
- Profile image

5.6 Garage Owner Screens

These screens are only available for garage owners.

5.6.1 Add Garage Screen

Purpose:
Allows garage owners to register their garage.

Fields:

- Garage name
- Address
- Location coordinates

- Services offered
- Phone number
- Garage images

5.6.2 Manage Garage Screen

Purpose:
Allows owners to update garage information.

Features:

- Edit services
- Update availability
- Update contact information

5.6.3 Mechanic Requests Screen

Purpose:
Displays mechanic service requests from users.

Information Displayed:

- User name
- Vehicle model
- Problem description
- Requested service

Actions:

- Accept request
- Reject request

6. Navigation Architecture

This section describes how users navigate through the QirbGarage mobile application. The navigation system is designed to provide **intuitive access to core modules while maintaining a clean and scalable structure.**

The application will use a combination of:

- **Bottom Navigation Bar (Primary Navigation)**
- **Stack Navigation (Screen-to-screen transitions)**
- **Modal Navigation (Temporary interactions such as forms)**

6.1 Navigation Structure Overview

The application navigation follows a **hierarchical structure** starting from the splash screen, authentication flow, and main application interface.

Navigation Flow Overview

Splash Screen
↓
Authentication Flow

(Login / Register / Role Selection)
↓
Main Application Interface
↓
Bottom Navigation Modules

6.2 Primary Navigation (Bottom Navigation Bar)

After authentication, users enter the **main application interface** which uses a **Bottom Navigation Bar** to switch between core modules.

Main Navigation Tabs

1. Find Garage (Home)
2. Central Spare
3. Manual Hub
4. Profile

Navigation Structure

Main Application

<u>Bottom Navigation</u>

Find Garage | Central Spare | Manual Hub | Profile

Purpose of Bottom Navigation:

- Provides quick access to core application features
- Maintains consistent navigation across the application
- Allows users to switch modules without losing context

Default Tab: **Find Garage**

6.3 Authentication Navigation Flow

Users must complete authentication before accessing the main application.

Navigation Flow:

Splash Screen
↓
Login Screen
↓
Register Screen (if user is new)
↓
Role Selection Screen
↓
Main Application

Behavior:

- Splash screen checks if the user is already authenticated
- If authenticated → navigate directly to Home screen
- If not authenticated → navigate to Login screen

6.4 Find Garage Navigation Flow

The Find Garage module allows users to search for garages and request services.

Navigation Flow:

Find Garage Screen
↓
Garage Details Screen
↓
Request Mechanic Screen
↓
Request Confirmation

Alternative Navigation

Find Garage Screen
↓
Garage Details Screen
↓
Garage Map Screen

Navigation Behavior:

- Selecting a garage card opens the **Garage Details Screen**
- Users can either **contact the garage** or **request a mechanic**
- Location option opens **Garage Map Screen**

6.5 Central Spare Navigation Flow

The Central Spare module functions as a marketplace for spare parts.

Navigation Flow:

Central Spare Feed
↓
Spare Part Details
↓
Contact Seller

Post Creation Flow

Central Spare Feed
↓
Create Spare Post Screen
↓
Post Confirmation

Part Request Flow

Central Spare Feed
↓

Request Spare Part Screen
↓
Request Published

Navigation Behavior:

- Users can browse spare parts through the feed
- Selecting a spare part opens the **details screen**
- Users can create posts for selling or requesting parts

6.6 Manual Hub Navigation Flow

The Manual Hub provides educational repair resources.

Navigation Flow:

Manual Hub Feed
↓
Manual Details Screen
↓
Manual Viewer (PDF / Video)

Video Learning Flow

Manual Hub Feed
↓
Video Manual Viewer
↓
Full Manual Details

Navigation Behavior:

- Manuals are categorized by type (engine, electrical, body, general)
- Users can view manual previews before accessing full content
- Premium manuals may require purchase or access permissions

6.7 Profile Navigation Flow

The profile module allows users to manage personal information and activity.

Navigation Flow:

Profile Screen
↓
Edit Profile Screen

Additional Navigation

Profile Screen
↓
My Spare Posts
↓
Post Details

Profile Screen
↓
My Manuals

6.8 Garage Owner Navigation Flow

Garage owners have additional navigation options related to garage management.

Navigation Flow

Profile
↓
Add Garage Screen

Garage Management Flow

Profile
↓
Manage Garage Screen
↓
Update Garage Details

Mechanic Requests Flow

Profile
↓
Mechanic Requests Screen
↓
Accept / Reject Request

Navigation Behavior:

- Garage owners can manage their garage listings
- Owners can review and respond to mechanic service requests

6.9 Navigation Types Used in the Application

The application uses different navigation patterns depending on the interaction.

Bottom Navigation

Used for switching between main modules.

Modules:

- Find Garage
- Central Spare
- Manual Hub
- Profile

Stack Navigation

Used for navigating between screens within a module.

Examples:

Garage List → Garage Details → Request Mechanic

Spare Feed → Spare Details → Contact Seller

Modal Navigation

Used for temporary interactions such as:

- Creating spare part posts
- Uploading manuals
- Submitting mechanic requests

Modal navigation prevents users from losing context in the main screen.

6.10 Navigation Design Considerations

To ensure a smooth user experience, the navigation architecture follows these principles:

- Consistent navigation across all modules
- Minimal number of steps to reach key actions
- Clear hierarchy between primary and secondary screens
- Fast switching between core modules using bottom navigation
- Logical screen transitions within each feature

6.11 Estimated Navigation Depth

The maximum navigation depth in the application is expected to be **three levels**.

Example:

Home (Garage List)
↓
Garage Details
↓
Request Mechanic

Limiting navigation depth improves usability and keeps interactions efficient.

7. Flutter Architecture

The mobile application will follow Clean Architecture principles.

Layers:

Presentation Layer

Contains UI screens, widgets, and BLoC state management.

Domain Layer

Contains entities, use cases, and repository contracts.

Data Layer

Handles models and data sources interacting with backend APIs.

8. Flutter Folder Structure

```
lib/
  core/
    constants
    utilities
    services
  features/
    authentication/
    garages/
    spare_parts/
    manuals/
```

Each feature contains:

```
  data/
  domain/
  presentation/
```

presentation includes:

```
  screens
  widgets
  bloc
```

9. State Management

State management will use the BLoC (Business Logic Component) pattern.

Benefits include:

- predictable state updates
- clear separation between UI and logic
- scalable architecture for enterprise apps
- easier testing and debugging

10. UI Design Principles

The UI design will prioritize:

- Clean and minimal layout
- Fast navigation
- Clear information hierarchy
- Mobile-first responsiveness
- Accessible components
- Reusable widgets

11. Location Services

Location services allow the application to detect the user's position and display garages nearby.

Features include:

- automatic location detection
- distance calculation
- nearby garage listing

12. Performance Considerations

The application should maintain high performance by:

- optimizing widget rebuilds
- using efficient BLoC state handling
- caching frequently used data
- lazy loading lists

13. Security Considerations

Although backend security is handled server-side, the frontend should:

- securely store tokens
- validate inputs
- protect sensitive user actions

14. Future Improvements

Potential improvements include:

- In-app chat
- Mechanic live tracking
- Booking system
- Online payments
- Push notifications